



59:51 Hledání ... Vyučující Katedry Třídy Klasifikace Prohlížení dle oborů/plánů Nastavení

Popis předmětu

Nevíte si rady s předměty? Možná Vám pomůžou naše [návod](#)y. **V případě problémů se ujistěte, že máte modul nastaven do správného akademického roku.**

Byla nasazená funkcionalita, která zajišťuje, že soubory nahrané k předmětu "ve vztahu matka" jsou dostupné u předmětů "ve vztahu dcera".

Pokud vám nejdou editovat texty u vašeho předmětu, zkontrolujte si, že máte nastavený správný akademický rok. Změnu můžete provést v menu nastavení.

Economics of Green Deal - JEB159

Anglický název: Economics of Green Deal

Český název: Economics of Green Deal

Podoba výuky: přednáška

Zajišťuje: [Institút ekonomických studií \(23-IES\)](#)Fakulta: [Fakulta sociálních věd](#)

Platnost: od 2025

Počet semestrů výuky: 1

Semestr: zimní

E-Kredity: 5

Způsob provedení zkoušky: zimní s.:písemná

Rozsah, examinace: zimní s.:2/0, Zk [\[HT\]](#)Počet míst: 59 / neurčen (neurčen) [?](#)

Maximální kapacita předmětu: neomezen

Minimální obsazenost: neomezen

4EU+: ne

Virtuální mobilita / počet míst pro virtuální mobilitu: ne

Stav předmětu: vyučován

Jazyk výuky: angličtina

Forma uskutečňování: prezenční

Možnost opakovaného zápisu: 2 / 2 / 1 / 2 [?](#)Poznámka: předmět je možno zapsat mimo plán
povolen pro zápis po webu
při zápisu přednost, je-li ve stud. plánuGarant: [Mgr. Milan Ščasný, Ph.D.](#)Vyučující: [Mgr. Vědunka Kopečná, Ph.D.](#)[JUDr. Vojtěch Máca, Ph.D.](#)[Mgr. Matěj Opatrný, Ph.D.](#)[Mgr. Lukáš Rečka, Ph.D.](#)[Mgr. Milan Ščasný, Ph.D.](#)[Mgr. Iñaki Alberto Veruete Villegas](#)Třída: [Courses for incoming students](#)Prerekvizity [?](#): JEB104[Zapsat](#) [Termíny zkoušek](#) [Rozvrh ZS](#) [Nástěnka](#)**Export:**typ: --- jazyk: [česky](#) [?](#) [Uložit](#)

Soubory	Komentář	Kdo přidal
ARTICLE_01a_Verhoef_2002.pdf		Mgr. Milan Ščasný, Ph.D.
ARTICLE_01b_Verhoef_1994.pdf		Mgr. Milan Ščasný, Ph.D.
ARTICLE_02_Kling_2025.pdf		Mgr. Milan Ščasný, Ph.D.
ARTICLE_04_Ch17_CBA.pdf	ARTICLE_04_CBA	Mgr. Milan Ščasný, Ph.D.
EGD_Week_9_Distributional_Impacts (Opatrný).pdf	Distribuční dopady - week 9	Mgr. Matěj Opatrný, Ph.D.
00_Intro.pdf	Lecture 01 Intro	Mgr. Milan Ščasný, Ph.D.
01_Regulation_Externality.pdf	02_Externality-Regulation	Mgr. Milan Ščasný, Ph.D.
03_EGD.pdf	03_EGD	Mgr. Milan Ščasný, Ph.D.
04_EGD-II.pptx	04_EGD-II	Mgr. Milan Ščasný, Ph.D.
06a_Energy Markets (Kubecek).pdf	06a_Energy System (Kubecek)	Mgr. Milan Ščasný, Ph.D.
06b_Energy system modelling (Recka).pdf	06b_Energy System (Recka)	Mgr. Milan Ščasný, Ph.D.
09_Modelling.pdf	09_ModelReview_Technology	Mgr. Milan Ščasný, Ph.D.

Anotace - angličtina

The European Green Deal is the key policy to transform the EU into a carbon-neutral, resource-efficient and competitive economy. This transition brings many challenges to the economy and society that need to be changed concerning technology used, behaviour and lifestyle. This course will address these challenges through a series of lectures that will present approaches to rigorously assessing the impacts of the transition on the economy, the environment, and people. It will rely on environmental economics complemented by models and theories from other fields in economics (energy-, transport-, behaviour-) and disciplines (energy system modelling, environmental sciences). The course will discuss several topics: regulation (externality, policy instruments) & environmental problems (pollution, climate change); energy transition and the decarbonisation pathway, with a particular focus on the energy system and transport modelling; Regulatory Impact Assessment | Benefit-Cost analysis, and the role of consumer behaviour and choices in the energy transition.

Aim of the course

The course will address the environmental problems and climate change from the economic perspective, discussing how optimal regulation is defined and how to rigorously quantify the economic, social and environmental impacts of policies. Special attention is paid to modelling decarbonisation scenarios and pathways, including the role of technology and social innovation.

Poslední úprava: Ščasný Milan, Mgr., Ph.D. (29.09.2025)

Podmínky zakončení předmětu - angličtina

Active participation in the lectures [10 points]

Writing test with a satisfactory grade [65 points]

Open-book (all, but electronic) exam (written) – test understanding and application, not memorisation [35 points]

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Metody výuky - angličtina

Presentations with slides

Poslední úprava: Ščasný Milan, Mgr., Ph.D. (29.09.2025)

Kontroly studia předmětu a podmínky pro jejich úspěšné vykonání, způsob hodnocení - angličtina**Course completion requirements**

- Active participation in the lectures [10 points]
- Writing test with a satisfactory grade [65 points]
- Open-book (all, but electronic) exam (written) – test understanding and application, not memorisation [35 points]

Grading:

91 and more	A
81 to 90	B
71 to 80	C
61 to 70	D
51 to 60	E
50 and less	Failed

Poslední úprava: Ščasný Milan, Mgr., Ph.D. (29.09.2025)

Syllabus - angličtina

The course consists of the following lectures.

Module 1: Foundations and Policy (Weeks 1-4)**Introduce the Course [Milan Ščasný]****Week 1 [September 29] Setting the Stage – Climate Change: Physical Foundation and Policy Implications [Miroslav Havranek, CENIA & Charles University Environment Center]**

- From Mitigation to Adaptation to Global perspective and Geoengineering solutions
- Historical context: From Kyoto to Paris to the Green Deal

Week 2 [October 6]: Environmental Externality and Regulation [Milan Ščasný]

- **Environmental externalities** - Start with a coal power plant: it produces electricity (private good) but also CO2 and air pollution (social costs not reflected in price). Show how this market failure leads to overproduction of emissions and why government intervention is economically justified, using simple graphs with social vs private costs.
- Optimal externality: quantification, optimal level; internalisation: Coase vs Pigou
- **Policy instruments: Carbon tax vs ETS vs regulations** - Compare three ways to fix the externality: carbon tax (set the price, let quantity adjust), ETS (set quantity, let price adjust), and regulations (efficiency standards). Each has trade-offs: efficiency, effectiveness, acceptability.
- **Static and dynamic efficiency** - Static efficiency asks "what's the cheapest way to cut emissions today?" Dynamic efficiency asks "what creates incentives for innovation tomorrow?"

Week 3-4 [October 13 & 20]: European Green Deal

[Richard Jurik, Ministry of the Environment & Vojtech Maca, Charles University Environment Center]

- **Key components and 2030/2050 targets**—The Green Deal sets legally binding targets: 55% emission reduction by 2030 and net zero by 2050. But it's more than climate policy. It includes a circular economy (reuse/recycle), biodiversity strategy, and zero pollution ambition—it is not just environmental policy.
- **Fit-for-55 package breakdown** - strengthening existing tools (EU ETS), creating new ones (ETS2, CBAM), and updating standards (cars, renewable energy). The key insight is policy coherence - all pieces must work together or carbon just "leaks" between sectors or countries.
- **Focus on: EU ETS reform, ETS2 (buildings/transport), CBAM (carbon border tax)** prevents European industry from relocating to avoid carbon costs.
- **Revenue recycling and funding mechanisms** - The €1 trillion Green Deal funding sounds impressive but break it down: mostly redirected existing funds plus private investment "mobilized" by public spending. The new Social Climate Fund will compensate vulnerable households.
- **What are NECPs and why they matter** - National Energy and Climate Plans as an 10-year strategy each member state must submit, showing how they'll meet EU targets using their specific energy mix and economic structure.
- **Czech NECP development** - Czech Republic faces unique challenges: heavy industry (steel, chemicals), coal dependency (40% of electricity), and nuclear debates (expand Dukovany/Temelin or not?).

Week 5 [The Dean's Day]

Module 2: Climate and Energy System modelling (Weeks 6-8)**Week 6 [November 3]: Climate Economics and Social Cost of Carbon [Milan Ščasný]**

- What climate change is about & Why does it matter? How temperature increases translate to economic losses through agriculture, health, labour productivity, and extreme events.
- **Integrated Assessment Modelling** and its key parameters: **Social Cost of Carbon**—how to compute/estimate it; Social Discount rates & intergenerational equity—the key ethical question: how much should we value future generations' welfare versus current costs?
- **How the countries use SCC in policymaking—Unlike the US**, where SCC is central to policy evaluation, the EU uses it more as a benchmark to check if carbon prices are "right." Show how the EU ETS price converging toward SCC estimates gives confidence that the market mechanism is working, but also why some sectors need additional policies.

Week 7-8 [November 10 & 17]: Energy System Transformation

[Vladimir Kubecek, Czech Transition System Operator & Lukas Recka, Charles University Environment Center]

- **EU energy markets and challenges** - Europe's electricity market was designed for fossil fuel plants that can turn on/off as needed, but renewables produce when the sun shines or wind blows, not when we need power. This mismatch creates price volatility and requires fundamental market redesign plus massive grid investments.
- **Renewable integration and merit order effects** - The merit order means cheapest plants produce first; but more renewables mean lower prices, which reduces their profitability and discourages investment - hence the need for subsidies and capacity payments; load curve vs duck curve; flexibility (supply and demand), smart grids; dispatchable and non-dispatchable generation, transmission & distribution, storage
- **Energy security vs climate goals** - The Ukraine crisis exposed Europe's vulnerability: depending on Russian gas while trying to decarbonize was always contradictory. **REPowerEU** and recent energy crisis lessons - **REPowerEU** accelerated renewable deployment not for climate but security reasons, showing how geopolitics can accelerate or derail climate policy.
- Energy system modelling vs. Power dispatch modelling: Energy system optimisation models (TIMES) vs. Dispatch and adequacy

Module 3: Tools and Analysis (Weeks 9-12)**Week 9 [November 24]: Economic Impacts and Trade-offs [Vedunka Kopečna & Inaki Veruette, Charles University Environment Center & IES]**

- Approaches: Input-output analysis; Economic models (neoclassical CGE models vs. post-Keynesian models)
- **Neoclassical critics:** emphasise real-world complexities such as uncertainty, financial instability, and the role of effective demand. **Hybrid models** combine various frameworks to address limitations in traditional models.
- **Macroeconomic feedback loops** - Carbon pricing raises energy costs, increasing inflation, potentially triggering interest rate rises, reducing investment - but green investment creates jobs, boosting demand, potentially overheating the economy. These feedback loops mean climate policy is now macroeconomic policy, requiring coordination with central banks and finance ministries that students should understand
- **E3ME for economy-wide impacts** - E3ME is a macro-econometric model that captures real-world rigidities: unemployed workers don't instantly find green jobs, factories can't immediately switch technologies, and investments take time to materialise. Show how carbon pricing initially reduces GDP but green investments create stimulus - but timing and distribution matter enormously.

Week 10 [December 1]: Distributional Impacts and Energy Poverty [Matej Opatrný, Charles University Environment Center & IES]

- **Who pays for the transition?** - Carbon pricing is regressive: poor households spend 10% of income on energy versus 3% for rich households, so uniform carbon prices hit the poor harder. But the rich emit more absolutely, so they pay more in euros
- **Ex post energy and transportation poverty** – how poverty is measured? Pros and cons of conventional indicators
- **Ex ante social impact assessment** - Walk through actual microsimulation by the DASMOD model
- **Just transition:** Social Climate Fund and compensation mechanisms

Week 11 [December 8]: Behavioral Aspects [Milan Ščasný]

- **Behavioral responses to climate policies** - People don't respond to carbon prices as economics textbooks predict: they undervalue future energy savings (myopia), stick with defaults (inertia), and follow others (social proof); consumers choice on low-carbon technologies; consumer demand and energy pricing.
- Energy savings & energy efficiency paradox, rebound effect; nudging & behav models
- **The role of technology vs behavior change** – how they are incorporated in the macro impact assessment models

Week 12 [December 15]: How Economists Model the Transition: Synthesis and Critical Assessment [Milan Ščasný]

- **Overview of modeling approaches: synthesis** - Models range from engineering optimization (what technology mix minimizes costs?) to economic equilibrium (how do prices adjust across the economy?) to sectoral detail (hourly electricity market simulation). Each answers different questions and makes different assumptions
- **Dealing with uncertainty** - models can't predict technology breakthroughs (fusion), geopolitical shocks (Ukraine war), or social tipping points (flight shame). Show how analysts use scenarios, sensitivity analysis, and robust strategies to handle deep uncertainty.
- **Carbon leakage and global cooperation** – carbon footprints and trade
- **Emerging challenges and course synthesis** - New challenges emerge constantly

Poslední úprava: Ščasný Milan, Mgr., Ph.D. (29.09.2025)

**Statistika**

Akademický rok	semestr	# předběžně zapsaných	# v rozvrhu	# závazně zapsaných bez uznaných	# splněných (vice)	% úspěšnost	průměrná známka ?	# zkoušení	průměrný úspěšný termín
2025/2026	zimní	31	29	29	24	82.8	1,96	28	1,08
	letní	-	-	-	-	-	-	-	-
2024/2025	zimní	40	40	40	35	87.5	1,8	37	1
	letní	-	-	-	-	-	-	-	-
2023/2024	zimní	-	-	-	-	-	-	-	-
	letní	-	-	-	-	-	-	-	-

2022/2023	zimní	-	-	-	-	-	-	-	-
	letní	-	-	-	-	-	-	-	-
2021/2022	zimní	-	-	-	-	-	-	-	-
	letní	-	-	-	-	-	-	-	-